

0.前置

$$\omega_n = \begin{cases} \min\{\alpha \mid \alpha \in Ord \wedge \|\alpha\| = \aleph_n\} & n > 0 \\ 1 & n = 0 \end{cases}$$

$$\forall \alpha \in On \forall A (\forall \beta \exists \gamma (\beta, \gamma \in A \leftrightarrow \beta \in \gamma)) \leftrightarrow A \text{ aft } \alpha = \{\mu : \alpha \in \mu, \mu \in A\}$$

$$A = (a_1, \dots, a_n) (a_i \in On, 1 \leq i \leq n):$$

$$\bullet (n = 1 \rightarrow B = \Pi_{a_1}) \vee (n > 1 \wedge ((a_1 \leq a_2 \rightarrow B = \Pi_{a_n} \text{ onto } seqturnrefl((a_2, \dots, a_n))) \vee (a_1 > a_2 \rightarrow B = \Pi_{a_n} \cap seqturnrefl((a_2, \dots, a_n)))) \leftrightarrow seqturnrefl(A) = B$$

$$\bullet gp(A) = \begin{cases} (0) & n \leq 1 \\ (a_{i+1}, \dots, a_n) & \exists i (i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j (j \in i \wedge a_j < a_1)) \\ (a_2, \dots, a_n) & \neg \exists i (i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j (j \in i \wedge a_j < a_1)) \end{cases}$$

$$\bullet bp(A) = \begin{cases} (a_1 - 1) & n \leq 1 \\ (a_2, \dots, a_i, a_1 - 1) & \exists i (i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j (j \in i \wedge a_j < a_1)) \\ (a_2, \dots, a_n, a_1 - 1) & \neg \exists i (i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j (j \in i \wedge a_j < a_1)) \end{cases}$$

$$(A \text{ onto})^\alpha B = \begin{cases} A \text{ onto } (A \text{ onto})^\beta B & \alpha = \beta + 1 \\ \{\mu \mid \forall \beta < \alpha, \mu \in (A \text{ onto})^\beta B\} \end{cases}$$

$$\alpha \text{ th } A = \begin{cases} \sup\{\nu : \mu \text{ th } A, \mu \in \alpha\} & \alpha > 0 \\ 1 & \alpha = 0 \end{cases}$$

$$(A \text{ onto})^\alpha B = \begin{cases} A \text{ onto } (A \text{ onto})^\beta B & \alpha = \beta + 1 \\ \{\beta \mid \forall \nu < \alpha, \exists \beta \in (A \text{ onto})^\nu B\} & cf(\alpha) = \omega \\ \{\nu \mid \forall \beta < \nu, \nu \in (A \text{ onto})^{\alpha[\beta]} B\} & cf(\alpha) = \omega_1 \end{cases}$$

1. MOCF

$$C(\alpha, \beta, 0) = \beta$$

$$C(\alpha, \beta, n + 1) = \{\nu + \mu, \nu * \mu, \nu^\mu, \psi_\xi(\gamma), \Omega_\nu, \mathbf{I}_\nu : \gamma, \nu, \mu, \xi \in C(\alpha, \beta, n) \wedge \gamma \in \alpha \cap C(\gamma, \xi)\}$$

$$C(\alpha, \beta) = \bigcup_{i \in \omega} C(\alpha, \beta, i)$$

$$\psi_\pi(\alpha) =$$

$$\begin{cases} \min\{\nu : \nu \in ((seqturnrefl(bp(B)) \text{ onto})^{\omega_1} seqturnrefl(gp(B))) / (\min(((seqturnrefl(bp(B)) \text{ onto})^{\omega_1} seqturnrefl(gp(B))) \text{ aft } \xi \text{ th } A) \\ \min\{\nu : \nu \in \pi \cap C(\alpha, \beta)\} \\ \dots \end{cases}$$